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Data Stories for Children and Older Adults: The German Energy Grid as a Case Study

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Abstract

The role of data visualisations in communicating information is becoming increasingly important as more data becomes available. As their popularity grows, the range of people who come into contact with visualisations expands as well. Therefore, not every visualization suits every user equally well. They differ widely in their prior knowledge of visualisations and the context of the data presented. A representation designed for domain experts with background knowledge in data visualisation can quickly become overwhelming and confusing for the general public, while simple visualisations may lack the detail needed to provide meaningful insights. People also have varying abilities to perceive and interact with data visualizations. For example, disabilities, experience with technology, mental capacity, or age are all factors that affect how a visualisation should be designed to convey information as effectively as possible. In this thesis, the aspect of age will be examined in more detail by looking at design principles and guidelines that can be followed to tailor data stories to children and older adults. For these target groups, design proposals for a data story will be made using the German energy grid as the content example.

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1 | Introduction

Over the past years, there has been a sharp rise in the use of data visualizations. They make it possible to present complex data in an understandable way with the goal to gain insights into the data or communicate findings to an audience. However, not every group can benefit equally from the advantages of visualizations because of varying circumstances, such as prior knowledge, experience with visualizations, age, or other factors. This means that a data visualization may be appealing to one person but incomprehensible to another. To ensure that as many people as possible can be informed through data visualizations, it is important to know how they should be designed to reach identified target audiences. To add to this, the ability to properly interpret visualizations is becoming increasingly necessary to understand issues based on actual data, which is the foundation for forming well-informed opinions and making informed decisions in everyday life.

Since most people who consume media featuring data visualizations are adults, this age group is most frequently taken into account during the creation process of data stories. This is also reflected in the research on the effectiveness of visualizations. However, in a time of demographic change, it is becoming more important to consider a growing number of older people and provide important information in an age-appropriate manner. Keeping older adults well informed about data-driven topics can be challenging, particularly when they lack the technical expertise that is often required to make optimal use of visualizations or when they can't fully engage with them due to cognitive or sensory limitations. But it is still crucial that they stay up to date with insights that can be gained from utilizing data visualizations. Therefore, this problem should be a more widespread concern, and more research is needed on how data stories can be designed to help older people stay informed about significant topics.

If we take a look at the other end of the age spectrum, there is also a growing need to introduce children to the concepts of data visualization early on. This is especially important because children get confronted with them regularly in everyday life, and visualizations can be misleading when not read correctly. However, not enough is being done to equip children with the skills to interpret and understand data visualizations in school. So there is a need to broaden data visualization literacy among young children. Again, not as much research has been conducted on this age group compared to adults, also partly due to the challenges of conducting studies with children. So more research needs to be done

not only on how children can learn to draw conclusions from presented visualizations, but also on effective teaching methods and tools to improve their visualization literacy.

An area where visualizations are frequently used is energy production in Germany, as there is a great deal of complex data available on this topic that is nearly impossible to fully understand without visual aid. Furthermore, it affects nearly the entire population in their daily lives and beyond, which is why there is significant media and political interest in this topic. It is therefore important to provide information based on reliable data to as many people as possible, not least because individual actions can have a major impact on the overall electricity system. Therefore a variety of visualizations with different focuses and varying levels of complexity are required in order to reach differing target audiences who may have trouble fully grasping visualizations that are currently widely used in media.

This thesis explores how data visualizations can be designed to be more accessible and understandable for older adults and children using the German energy grid as an example. This is done by reviewing design principles and guidelines that have been proposed in prior research to derive a set of specific design recommendations for each target group. Furthermore, current interactive data visualizations of the German energy grid on the internet will be analyzed to see which problems might arise for the respective target groups when using them. Based on the findings of the literature review and the analysis of visualizations two data stories about German electricity generation are designed and implemented. One targeted towards older adults and one for children, turning the theoretical design principles into interactive visualizations.

2 | Theoretical Background

In this chapter, the findings of previous research on the topic of visualizations for different target groups shall be explored. Design guidelines and principles for older adults and children will be examined in more detail.

The field of data visualization has steadily gained relevance in recent years, as multiple studies have shown that visualizations can be effective at delivering information to an audience (e.g. [Pan+14], [Kna25]). However, to what extent this effect occurs is still debated, as not every visualization is equally effective depending on the quality and the context ([Haa+18]). With growing popularity, many methods and principles of representing data more visually have been developed over the last few years. But a research topic that is still underexplored is how data visualizations can be tailored to fit the needs and requirements of a specific group of people. It is essential to know the intended target audience when creating visualizations. Because the conclusions people draw from them can vary a lot depending on a person's background knowledge, age, their prior opinion, context of the visualization and many more factors.

One of the most important aspects to consider is the visualization literacy of the target audience. In [Boy+14], visualization literacy is defined as

the ability to confidently use a given data visualization to translate questions specified in the data domain into visual queries in the visual domain, as well as interpreting visual patterns in the visual domain as properties in the data domain.

There are some other related terms and definitions, such as visual literacy, which refers to the ability to understand, interpret and evaluate visual messages in general. But visualization literacy is the term most commonly used in the context of data visualizations.

The concept of visualization literacy is mentioned in several studies. In [Boy+14], a method is developed to assess the visualization literacy of individuals by presenting them with various tasks on different types of data visualizations. They use a very structured approach in order to develop a set of tasks that can reliably measure the visualization literacy of individuals. [Bör+16] uses a more empirical method by observing if people have seen certain representations before and how they would interpret the underlying data. They tested different kinds of charts, graphs, maps and network layouts. The key takeaway from this study is that most people only recognized basic data visualizations like charts and graphs

and struggled to make sense of more complex visualizations. It is also worth noting that the study was conducted in a science museum, which might have led to better results than what is expected from the general population. This hints towards a gap between the visualization literacy of regular people and the expectations of visualization researchers and creators. Therefore, [Bör+16] makes a call for more visualization education of the general public to close this gap.

[Rus+07] already noticed this need for improving visualization literacy in 2007. They propose workshops for students not only in computer science but also in other disciplines, since visualization are used across various research fields and in a wide range of workplaces. As a result of this, they also highlight that improving visualization literacy through education has to be a collaborative effort involving multiple areas of expertise like computer scientists, designers, educators and domain experts. Since then, more approaches to visualization education have been developed. As an example, [KSK24] explores the possibility of using games to advance data visualization literacy. For this study, they use the existing games Townscaper and Stardew Valley to teach players about data visualizations, as they have to interpret and manipulate visual information to achieve goals and progress in the game. This approach seems promising as it combines a fun and engaging activity with educational content. They also see potential in games, especially those made for improving visualization literacy. However, it requires some effort to design and develop such games.

How a person interprets a visualization not only depends on their visualization literacy but also on their prior knowledge of the subject. For example, [Váz+24] compares experts in the field of data visualization to PhD students, showing that people from different backgrounds focus on varying things when confronted with a visualization. While visualization experts focused on technical aspects and design choices, the non-experts paid more attention to the content and relied more on textual elements. They were also more likely to misinterpret a visualization when not guided through it. Considering the subject knowledge of the intended target audience is important in order to include the right amount of detail in the content of the visualization. This is a rather difficult balance of not overwhelming users while also keeping them interested and presenting new insights. Since this issue is very dependent on the exact context and topic of a visualization, finding general theoretical guidelines in this regard is complicated.

Another aspect that is often overlooked when creating data visualizations is accessibility. With the growing popularity of visualizations, naturally more people with disabilities come in contact with them. So the question has to be asked: how can people with different disabilities perceive them? [Mar+21] discusses the challenges of this topic and gives advice on how to make a visualization more accessible. They make a distinction between different types of impairments: visual impairments, cognitive and learning difficulties and motor impairments. There are already existing standards for accessibility issues for general web content like the Web Content Accessibility Guidelines (WCAG) by the W3C, but more

research needs to be done on how to implement them into data visualizations. They also note that accessibility research in general can be quite challenging and costly, as there is a limited pool of candidates for participation in studies and the needs of disabled people can vary extensively for different individuals. Also, there is not a lot of monetary incentive to fund accessibility research.

While [Mar+21] examines a wider range of disabilities, [Kim+21] focuses on visually impaired people since most visualizations, especially in everyday situations, heavily rely on sight. Both [Kim+21] and [Mar+21] believe that there should be a widespread push for more awareness regarding these problems. Not only in terms of research on this topic, but especially among those who create data visualizations of any kind to reach a large audience. However, they both acknowledge that creating inclusive visualizations is often only possible within limits of available tools, resources and knowledge.

[EBM22] provides a framework for assessing the accessibility of a visualization by considering seven principles of inclusive design. Along with the four principles defined in the Web Content Accessibility Guidelines, which contain advice on how to make content Perceivable, Operable, Understandable, and Robust, they add three more: Compromising, Assistive, and Flexible. Some of these guidelines not only help improve the accessibility of a visualization but can also be used as basic guidelines for any visualization.

Another essential factor for the success of data visualizations is the story around it. This is also the point where a data visualization becomes a data story. [Nei+20] argues that the best way of communicating scientific findings is through storytelling. They do a systematic literature review on the topic and highlight eight aspects to make data storytelling effective. They include reducing complexity, focusing on a main message, picking the right data, structuring the story and knowing your audience. Especially knowing the audience is crucial, as it affects most of the other elements of effective data storytelling.

Data stories are not only used in scientific communication but also in various other fields such as business analytics, marketing or journalism. Some influential works on data storytelling include books like [Kna25] and [Dyk19]. As the titles of these books suggest, [Kna25] focuses more on storytelling with data, meaning that the story is supported and enhanced by the data. On the other hand, [Dyk19] is more about data storytelling, where the data is the main focus and the narrative is built around it. This is an important distinction that should be considered when combining a story with data.

While it is widely acknowledged that storytelling can add valuable context to data visualizations, there is also some scepticism. For example, [BDF15] notices that telling a story should not hinder the exploration of an interactive visualization. They argue that telling a too extensive story around a visualization might lose people's attention, especially in fast-paced environments like the internet. The main argument is that after reading the story, users might not engage with the visualization

itself as deeply as they would have otherwise. So there needs to be a balance between providing context and allowing users to explore the visualization independently.

When it comes to visualizing data related to the energy grid, various approaches have been developed. Since this is quite a broad topic, there are visualizations for many aspects of the energy grid, including generation, transmission, distribution, and consumption.

[FK22] did a survey on visualization methods for power grids, comparing different proposed approaches. Most traditional visualization methods that have been used in practice for decades are typically geographical and topological representations and time-series bar charts in separate views. They usually only provide a visualization of a steady state of the power grid rather than capturing dynamic changes over time. This becomes a challenge with the rising complexity of the power grid due to decentralized generation, fluctuations and smart grids. Some approaches differing from the rather simple industry standard, such as [Won+12], [Won+09] or [OW00], aim to address these challenges by adding elements like animations, virtual environments, line flows and more. However, combining different types of data, like geographical information, real-time measurements and simulations, into a coherent visualization is difficult.

Another modern approach is presented by [CCS23]. They developed *Grid-Viz*¹, a JavaScript library that can be used for creating web-based interactive visualizations of the energy grid. The data used includes transmission lines, substations, maps, and prices. They also note the challenge of dealing with the multi-modal nature of the data, especially the interactive combination of geographical information and real-time measurements.

In these visualizations, there is always a trade-off between the level of detail and the resulting complexity of the visualization, which might come with usability issues. Furthermore, adopting new approaches in a real-world scenario can be challenging due to factors such as existing infrastructure and user familiarity, which partly explains the limited research in this area. Another reason for this lack of research is that a lot of the data needed for these visualizations is not readily available to the public, because publishing it reveals sensitive information about critical infrastructure, which brings up security and privacy concerns.

These approaches are made with the goal of providing detailed and insightful representations of energy grid-related data to experts in the field, such as engineers, researchers, and policymakers. Without background knowledge, these visualizations are difficult to understand for a general audience. Some simpler visualizations of energy grid-related data designed for a broader audience in news and web media will be analyzed in chapter 3.

The aspect that this thesis will mainly focus on is the age of the target audience. In [Bör+16], the familiarity of different forms of visualizations is tested between youth and adults. They come to the conclusion that, naturally, younger people do not recognize as many types of visualizations

¹<https://eurostat.github.io/gridviz/>

as adults. It also shows that simpler representations like charts, maps, or graphs are generally more recognisable and understandable. At the same time, the message in more complex designs might not be easy to understand. This study also shows that creating and using new forms of visualizations can be effective for scientific purposes. Still, when it comes to communicating to a wider audience, simplicity is key to making the visualization more understandable for most people.

2.1 Older Adults

The first target group, which will be examined in more detail, mainly includes older people aged 60 and above who have little to no experience with technology in general and have not come into contact with many data visualizations. In some research (e.g. [Whi25]), this group is called *people in late adulthood (PLA)*. This abbreviation will be used throughout this thesis. The age of 60 is used in multiple papers to define the beginning of late adulthood ([Whi25], [Le+18], [McC+18]). Of course, this age limit is not strictly applicable, as some people below 60 can fit into this group, while some older people might not face the problems that usually come with older age. Key challenges for this demographic include deteriorating cognitive and physical abilities, lacking digital literacy, and limited experience with modern visualizations. This chapter discusses how these factors affect their ability to use visualizations and what can be done to tailor them to their needs.

[Whi25] did an extensive analysis on data visualization for elderly people and summarises some of the current research. They cite results from [Pri+16] showing that colour encoding can reduce mental workload, which is especially important for PLA. However, colours should be easily distinguishable with high contrast and different hue values. As an example for this problem, [Whi25] notes that distinguishing gradient colour maps becomes increasingly difficult with age, which is also shown in other studies. For instance, the gradient map shown in Figure 2.1 from [Zei26] could be difficult to read for people with poor eyesight, especially against a white background. The issue of background colour is also investigated in [Whi25]. The question is whether PLA prefer positive or negative contrast polarity when using visualizations, more commonly known as light mode or dark mode. The results showed that no clear case can be made for either when looking at how accurately and quickly PLAs use visualizations. The same holds for younger adults. Other research, however, has shown that positive contrast results in better reading performance, hinting at a different impact of the contrast polarity on reading or using a visualization. But when only considering visualizations, most people in this study performed better with the contrast mode they already preferred beforehand. So it is always a good option to let the user choose if they want to use light or dark mode.

Similarly, font sizes for labels and other text are increasingly important to consider when designing visualizations for people who may have difficulty with reading small text. Some simple solutions include using larger

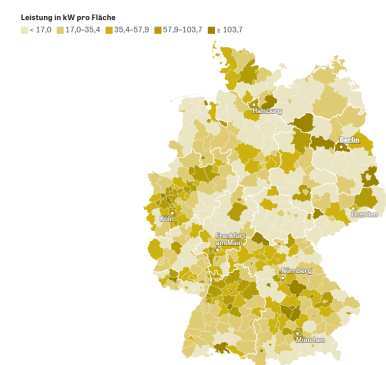


Figure 2.1: Colour gradient map used in [Zei26]

fonts or adding accessibility features like magnification or screen readers. To reduce these potential accessibility issues, it helps to consider existing tips for improving accessibility, such as the previously mentioned Web Content Accessibility Guidelines, when designing visualizations for PLA. The aim of these guidelines is to make any web content accessible, especially for people with any kinds of disabilities. The four key principles are to make content perceivable, operable, understandable and robust. These guidelines are quite extensive, since they include tips when creating any web content, so not all of it is applicable to visualizations. [Fan+23] also confirms accessibility issues when it comes to reading and adds that motor skills may also be a problem when it comes to precisely moving a mouse or using a touchscreen. These issues increase with smaller screens like phones or smartwatches, leading to the suggestion of using bigger screens like tablets, laptops or monitors to present visualizations to elderly people.

Furthermore, [Whi25] notes that PLA prefer familiar and simple visualizations. This may be due not only to potentially declining cognitive capacities but also to a lack of visualization literacy. However, it is uncertain which types of visualizations are perceived as familiar, as familiarity is often subjective. However, there are indications that they can work well with visualizations following the commonly used *overview-first, details on demand* principle, since they tend to prefer summarised data first and only want details when trying to explain a pattern. This is most likely also linked to reducing the cognitive load when using visualizations. Another inconclusive aspect is how much guidance to provide to help elderly people reach the intended conclusions. It has been shown that guiding elements such as explanatory text, visual aids that highlight important data points or adding explanation of a tool, help to improve understanding of a visualization among PLA ([CY24], [Whi25]). But there is a limit at which additional guidance can be perceived as visual clutter, making the visualization more confusing. This limit is difficult to define, and little research has addressed it so far. As [Whi25] puts it:

Balancing the tension between reducing clutter and adding scaffolding is an interesting open challenge in the context of visualization design for PLA.

Again, [Fan+23] makes similar considerations about visualization types and the amount of guidance. They recommend designing visualizations in a way that matches existing knowledge and experience to reduce mental workload, but note the individuality of what people expect from a visualization as well. [Fan+23] also notes the difficulty of providing enough help without causing exhaustion through visual clutter. As a possible solution, they suggest instructions on-demand, the option to only provide guidance when the user does not know what to do. Although further research is needed on this matter. They also advocate for a trial-and-error type of exploration when using interactive visualizations, with concise feedback on their actions, error messages and an easily recoverable home state they can return to if an unwanted interaction was taken. This is meant to help them explore a visualization without worrying about making a mistake due to inexperience with technology.

In [CY24], the general effectiveness of data visualization for older adults is tested with the example of flu vaccination. They test dashboards with various modifications to see which type older people prefer. The variables tested for are static or interactive, tailored or non-tailored, and with or without explanatory text. As a measure, they assessed perceived susceptibility to vaccination before and after seeing the dashboard. The overall score when presented with a visualization was higher than the control group, which means that they generally have a positive effect on informing elderly people. The interactive dashboards, which are becoming more widespread recently, seemed to deter or confuse seniors in this study, resulting in worse information recall when compared to the static representations. Whether interactivity in general leads to better results is still debated in research papers, with different studies coming to different conclusions. For example, [ZDW11] warns that interactive dashboards can backfire when not designed correctly and notices the need for further research on which elements of interactivity lead to better results. [AWK11] also got mixed responses on whether interactive visuals help in improving risk perception. The positive effect of explanatory text increased with age, indicating that descriptive text helps older people remember the information presented. Adding additional text with contextual information was also shown to be helpful by [Le+18] in a study investigating how to inform PLA about critical health data.

In addition to PLA having lower visualization literacy than younger people in general, multiple studies also mention that lower technology literacy can be an issue. In [Le+18], some of the participants even preferred additional paper printouts of visualizations, because they were not confident navigating a digital interface. [Whi25] also mentions this issue several times, naming a few factors that are important for PLA to adapt to technology coming from different other papers. Some of the most relevant include usability, accessibility, and design consistency. Perceived value, potential to improve quality of life, convenience, and confidence in learning to use technology also help make the adoption of digital visualizations easier. There are already many attempts being made to include PLA in digital life. For example, [McC+18] investigates how technological literacy of seniors can be improved by digital storytelling and social media with specifically targeted workshops. They found that interest-driven media participation has shown promising results. This means that people were more open to using and learning technology new to them when they had the freedom to choose a topic they cared about and learned the advantages of technology in everyday situations. To apply this to the context of visualizations, it can be assumed that the data and the content of the visualization should be interesting to the individual. This is also a plausible explanation on why most of the research is focused on health data. This topic concerns a lot of seniors personally, since monitoring health becomes more important to them.

[Fel+24] examines the difference between younger and older individuals by presenting them with a simple bar chart with varying small design features. As data, they used a simple step tracking over the span of a few days. Some context of the data was provided to help participants make

sense of the charts. They tested aspects of tooltips, data labels, data load, visual load, grouped bar charts, and stacked bar charts. After presenting the charts, they asked about specific data points that can be retrieved from the chart. This required different levels of understanding of the chart. Some were very simple, like 'How many steps did you walk on date d?', while others required some more steps, for example 'Between which dates was there an increase/decrease in the number of steps taken at an intensity level i?'. They conclude that the scores to questions about the charts were not as significantly different between the age groups, except for the grouped bar chart. The response time of older people, however, was much higher, which confirmed the expectation that cognitive speed declines with older age. So what can be taken away from this study is that older people still understand simple graphs like bar charts almost as well as younger people. It just might take them longer to process. Yet, one thing that needs to be considered when looking at these results is that most of the participants in this study already had at least some experience with data visualizations. So it can be expected that the overall scores for the general population might be a bit lower.

For this age group, most research focuses on presenting health data, since this topic becomes increasingly important with age and can potentially save lives. But the resulting design recommendations are not specific to health and can be applied to other kinds of data as well. The most prominent design principles for PLA are to use simple and familiar visualizations, provide guidance and context, and consider accessibility issues like colour contrast, font size, and motor skills. It is also important to consider the individual needs of the user, as there is a lot of variation in abilities and preferences within this age group. Much research also focuses on general interface design and digital literacy for PLA. To fully utilize these insights in visualization design, further research is needed. [Whiz5] wants to establish the term *GeontroVis* for this field of research. They define it as

a sub-field of visualization, encompassing research and practice of data visualization that primarily focuses on people in late adulthood (PLA).

It is going to be interesting to see how this relatively new area of research will develop in the next few years.

2.2 Children

The next target group are children in school around the age of 10. Again, this age is not very strict, as children often develop at a different pace and their skill levels regarding data visualizations can vary quite substantially. Much of the research for this group focuses on visualization education rather than informing on a specific topic. The aim is to give children the ability to recognise and understand visualizations from an early age. This knowledge can then be applied to visualizations about any subject. There are some very early works on data visualization education for children,

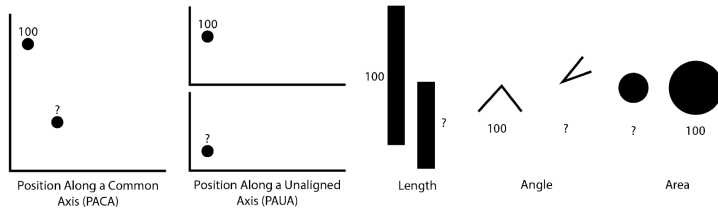


Figure 2.2: encodings tested for in [Pan+22]

like [Sch96] back in 1996. Nevertheless, the research in this specific area paused for a long time, since visualizations were not widespread at the time and the focus was mainly put on adults, who are the main consumers of data visualizations. In recent years, however, the need to teach children visualization literacy has increased significantly, as they are more likely to encounter visualizations at a young age. Because of this new research on this issue has emerged over the past years. Especially in combination with education becoming more digital, new options of visualization education are being developed. For example, [Alp+17] and [Bis+20] made tools meant for children to create their own simple visualizations.

[Pan+22] compares the graphical perception of children and adults by testing their accuracy on visualizations with varying design features. After reviewing their results, they conclude a few explicit design recommendations for visualizations targeted towards children. The first one is to use the same encoding rankings for children and adults, as the cognitive processing is similar, even though children make more mistakes on average. The ranking they give is from most accurate to least accurate: position along a common axis, length, position along an unaligned axis, angle and area. Examples of these encodings can be seen in Figure 2.2. The second recommendation they give is adding gridlines to improve the accuracy of children getting the correct value. Another piece of advice from this study is that the most accurate encoding is not always the best. They say that

Choosing a better encoding that provides slightly more accuracy may not be worth the loss in creativity or user enjoyment.

The last recommendation presented in this paper is to explore the data behind graphical perception guidelines further. While this is not advice specific to a single visualization, it is rather a recommendation for the entire field of data visualization to more accurately present findings of studies. The example they give is that in their study, the encodings of area and angle are similar in resulting accuracy.

[Alp+17] mentions that visualization literacy amongst the general public is relatively low, when taking into account the importance of visualizations in everyday life. They argue that an effective way to expand visualization literacy is to start at a young age in elementary school. They also make the point that teaching children effectively strongly intersects with other research areas like psychology and education in general. This research

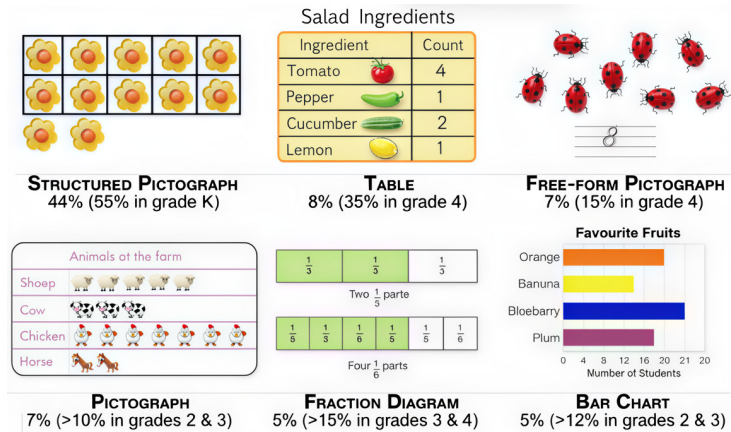


Figure 2.3: most common visualization types in textbooks by [Alp+17]

should also be considered when teaching children about data visualizations, for example, the Cambridge Handbook of teaching science ([22]) or research for digital educational interfaces ([Ovi13; Soti13]) to name a few. However, not much research has been done specifically on teaching visualization literacy, and most existing work focuses on adults, such as [Rus+07] or [HA17].

[Alp+17] also took a sample of existing textbooks in elementary schools and analyzed the most common types of visual representations. The results can be seen in Figure 2.3. In the same dataset, they analyzed the levels of abstraction from tangible objects to illustrations and spatial notations. In concurrence with education research, the level of abstraction increases as children get older. This is also called concreteness fading, going from a concrete physical model to a known form like a graphic or a photo, to a notation form like numbers, and is shown to be an effective learning method ([MF12]). This concept can easily be applied to visualization education. By using physical objects in lower grades, for example, something like the toolkit proposed by [Bae+23] and increasing the level of abstraction with familiar images like those in [Alp+17] or [Bis+20] used in their learning applications. With this approach, children can be slowly familiarised with most visualizations that use abstract numbers. [Alp+17] confirms that a large majority of teachers already use this theory. They also bring up three different kinds of exercises typically used for teaching children about data visualizations. These are reading exercises, where a given visualization is to be interpreted; completion exercises, like filling in or expanding a visualization template; and creation exercises, where the task is to create a visualization from scratch. Furthermore, they noticed that interactive tools are currently hard to find and that most of them do not support the possibility of children creating their own charts and are limited in their possibilities for teachers to customise the learning experience. To fill this gap, [Alp+17] developed a digital interactive tool for children to learn about visualization called *C'est la Vis*.

They set five design goals for developing the tool that can be widely expanded to visualizations for children in general. The first is to use multiple levels of abstraction, as already mentioned, so that teachers can

adjust the difficulty for different classes. The next goal closely related to this is to show the relationships between visualizations. In practice, this means animating or morphing abstraction levels into each other, which makes it easier to follow the steps of creating a visualization. The advantages of interacting with familiar things have already been discussed previously. This is the basis of the goal of direct manipulation, meaning the children can manipulate digital objects, preferably pictures of things in the real world, which is supposed to mimic physical interaction. Their next objective is to add variation to the type of tasks that teachers can create. The main tasks are reading, completing, and creating a chart, which are the most common types of exercises, as mentioned before. The last goal they introduce is to allow customisation of the visuals and the data. This is meant to add variety to the exercises teachers can create and to keep children engaged with multiple themes and contexts. The finished tool developed in [Alp+17] can be found at ².

[Bis+20] developed a similar tool called *Construct-A-Vis*. The focus of this paper is more on the process of children creating data visualizations rather than teaching how to read and understand them, as in [Alp+17]. So the goal for *Construct-A-Vis* is to build a tool mainly for creation exercises using free-form visualization creation. They use insights from general education research to set design principles for their application. For example, the principle of scaffolding and guidance means assisting the learning process with feedback from the application itself and from the teacher when used in a classroom setting. More guidelines include gradual skill development. The core idea is making tasks adjustable for different ages and skill levels of the children. This is also called configurability and is widely used in early education. In a visualization context, this can be seen in the concreteness fading approach, which [Alp+17] described in more detail. [Bis+20] implements this concept by providing the option to adjust colours, shapes, or size to fit various skill levels, while holding on to the free-form creation of visualizations. Another essential principle for [Bis+20] is to let children explore different visual mappings. They mention two different ways of mapping data to visuals. The first one, used by most popular tools, is the *data-driven visual mapping* approach, where data attributes are assigned to a visual. The other one, which is utilized in *Construct-A-Vis*, is called *visualization-driven mapping*. Here, visual elements are created first and then assigned to data. This is meant to help children explore the visual mapping more creatively while supporting the constructive visualization building approach, which is the fundamental idea of the tool. The last important design principle discussed in [Bis+20] is learning from peers and having shared interactions when creating and discussing visualizations. Some research has been dedicated to investigating collaborative visualization processes for adults, but not a lot for children in school, where interactions are important for children to learn together.

These design principles strongly influence the design choices made in the application. For example, the collaborative aspect is achieved by having an option to connect a third device to the personal devices of two children.

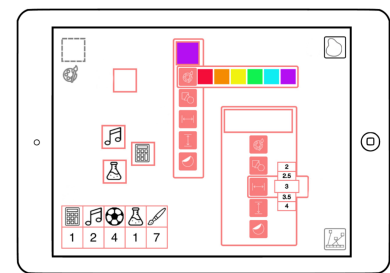


Figure 2.4: Sketch showing the functionalities of *Construct-A-Vis*

²<https://cestlavis.github.io/>

On this screen, both of them can place their created objects and create a visualization together. In a sample study, they found that limiting the feedback the application gives encourages the children to get feedback from each other and have discussions about their visualization. The configurability concept is implemented by having many different options for tasks that teachers can give to the children with varying difficulty. Tasks can include only one parameter, like the count of objects, or multiple, including colour, symbol, or shape of an object. By placing constraints on one or more of the variables, children have to get creative to figure out a way to represent the data. An example would be only allowing one object per data point instead of multiple, forcing children to adjust the size of the object. The fact that children come up with different charts for the same task is an indicator of creative processing, which is beneficial in the learning process. Another important finding that can be taken from the study is the need to guide children through the process of creating a visualization.

While the finished *Construct-A-Vis* tool is currently not open access, more information and a video on how it works can be found at ³.

Both of these programs use made-up datasets with familiar objects like animals, flowers, or everyday objects. This helps children imagine the data as an amount of objects, for example. [Bae+23] takes this a step further and recommends using datasets that the children can relate to personally or that they have even gathered themselves. This makes the insights into the data more interesting, since they better understand the data.



Figure 2.5: Assembled toolkit from [Bae+23]

[Bae+23] also proposes a physical toolkit children can interact with, so they can discover the basics of data visualization by guided play. They call this kit *Data Is Yours (DIY)*, and it can be assembled from a few physical components. The idea that physical interaction helps children with learning is widely supported in education research and is also mentioned in [Alp+17] and [Bis+20]. The DIY kit developed in [Bae+23] includes the means to create bar, line and pie charts. Children can make a paper template that can be recognised by a phone, and then the physical objects can be manipulated in a way that matches the template. Connecting the physical and digital world in this way helps children to interpret digital visualizations better. They show that this kit is quite effective in teaching children about data visualizations. Still, it is limited by the need for physical objects and educators to guide children in the right direction. This makes it difficult to implement in purely digital settings.

There are many other resources meant to help improve children's visualization literacy. Often, the concept of gamification is used to engage children by letting them play a game that teaches them about visualizations. The genre of educational games has been around since computer games became widespread in the early 2000s, and research has begun on their use in classroom settings ([Squo5], [Lee+04]). They have become more popular over the past few years, since the development costs of games in general have decreased a lot. Because of this, their effectiveness is being studied quite extensively. While the consensus is that games

³<https://haveacupofteawith.me/>

can definitely improve children's engagement, learning success seems to depend on multiple factors like game quality, appropriate usage, and how teachers utilize them ([KKK17], [BH13]). Some examples in the field of data visualization are *Diagram Safari*, developed in [Gäb+19], or an RPG-based game by [Huy+21]. The levels of gamification can vary greatly. For instance, the tools by [Alp+17] and [Bis+20], or the kit from [Bae+23] discussed previously, have some basic gamification but are still strongly focused on the graphs themselves without a greater story. The game by [Gäb+19] already has more gamification elements like fancier graphics, predefined levels that get increasingly harder and an underlying story. Their study has shown that performance in matching charts amongst the children has improved. However, they leave the question open as to how well solving in-game tasks can transfer to real-world problems. [Huy+21] takes the gamification even further with an RPG-type game. They introduce multiple characters, an open map for exploration, and a deeper underlying story that is told further as more tasks are completed. They argue that having a narrative increases engagement and enjoyment and can help to make abstract concepts of visualizations meaningful in the context of the game. All of these factors are supposed to lead to better learning outcomes. While the findings of their study cannot significantly prove that adding a narrative to the game leads to higher learning value, children had more fun and spent more time playing when guided by a narrative. They also confirmed that feedback from the game improved the learning experience, which is aligned with findings of other papers mentioned previously.

Another example of using a story in visualization education is the book called *Daphne Draws Data* ([Kna24]). It is written by Cole Nussbaumer Knaflc, who is also the author of *Storytelling With Data* ([Kna15]), a well-known data visualization guide for people in business, among other books on visualization targeted towards adults. In contrast to those books, *Daphne Draws Data* is specifically designed to improve the visualization literacy of children. This book is structured like a regular children's book, telling a story about a dragon that wants to learn about data visualizations, containing colourful illustrations and different characters. Children can help the main character, Daphne, solve problems by collecting data and using data visualizations to gain further insights. This is meant to teach them the benefits of data visualizations and give them an understanding of how to use them in a real-life context. Teachers or other educators can read the book to a group of children in a classroom or a dedicated workshop. They can guide them through the story and the visualizations while encouraging discussions and collaboration. Beyond the book itself, there are also videos, example tasks, and suggestions on how to best utilize the book on the project website at ⁴.

Although most of the discussed resources are mainly used to teach visualizations in general rather than to inform about a specific topic, they show effective methods to get children interested in a visualization, which is important before focusing on the visualization's content. Across the studies, researchers use rather simple datasets to avoid overwhelming

⁴<https://www.daphnedrawsdata.com/>

children with more data than they can comprehend. There is also the overarching theme of using data children can relate to and know from their own experience. So it is probably a good idea to follow this guideline when choosing the data for a visualization. Another important finding related to dataset simplicity is that all the papers limited the visualizations to basic types like bar charts, pie charts, or line graphs. This is another good recommendation when creating a data story for children, since they might not be familiar with more complex kinds of representations. A different concept present in almost every analyzed paper is interaction and giving children something to do. When implementing this, for example by giving them tasks to complete, it is important to provide helpful feedback. If possible, a teacher or an adult with some knowledge of visualizations can give feedback, but in any case the data story itself should provide guidance to help children reach the intended conclusions. More broadly, a visualization or data story should be colourful and visually interesting to catch and hold children's attention. Using familiar things like objects or characters can also improve engagement. Furthermore, the principle of a collaborative approach to data visualizations, through discussing or developing a data visualization, is mentioned quite often. It may be difficult to create the circumstances required, but it should be considered whenever possible.

3 | Analysis of current visualizations

In this chapter current visualizations in media shall be analyzed and the problems that might arise for the target groups when using them.

[ZZ16] analyzed the use of infographics in german newspapers in 2016 and the results show that most readers are generally interested in infographics. They found that most of the used visualizations are simple graphics like bar charts or maps. A problem they identified is that often the controls for interactive graphics are difficult to find, which leads to the interaction not being used as often. This can especially be the case for elderly people since they might not be familiar with common controls used in infographics, which aligns with the findings of research papers included in 2.1. A good example for this is the interaction of hovering to highlight specific data. This is found in a lot of visualizations across almost every website, even in the more simple graphics, e.g. in [Zeiz6] or [San26]. But the user has to find this themselves as there is no clear indication that this is a possibility. For more experienced technology users this might be no problem since hovering is a quite common interaction. But elderly people might not know this interaction when they have never seen it before. Therefore, a small text or info button could be added to explain the interactivity. To avoid visual cluttering all the possible interactions could be shown on demand with a small info icon, similar to what is used in some of the visualizations on [Zeiz6] or the information box before the section on energy prices from [RS26] to explain more context.

[ZZ16] divides infographics into different categories depending on their level of interactivity, course of action and communicative intent. Most of the visualizations in this study had low interactivity, with only some features like selecting specific data or a start button for an animation. These type of graphics can be found in most typical news stories like [San26]. Slightly fewer infographics had medium interactivity, allowing the user to manipulate a timeline or menu items letting them compare information. Only about 20% had high interactivity with the possibility of exploration using interactions like filtering options or inputs. Some examples of high interactivity graphics regarding the energy grid are most of the charts found on [Gan26]. The course of action of a visualization is divided into linear and non-linear. Linear means that there is a step-by-step predetermined path that the user follows through the interaction as intended by the author. This is also called author-driven,

since the reader has no real way to explore the graphic by themselves. The possibilities for the user with this type are limited to interactions like forward/backward or start/stop. This type only represent a small amount of the analyzed graphics in [ZZ16]. This is in part because this author-driven way of narration can also be achieved by using multiple static graphs in a particular order with explaining text telling a story about them. The largest part of visualizations are nonlinear, also called reader-driven. They allow free interactions like selecting or filtering data or changing the visualization, for example a choice between a bar and pie chart representation. Most of the graphics regarding the electricity grid found in online media provide at least a simple kind of nonlinear interaction like filtering data by year or energy source or highlighting specific data (e.g. in [Ze126]). There are also some infographics that use a mix of both techniques making them linear-nonlinear, where the author provides a path to follow but still gives the option for the reader to explore, e.g. with interactive timelines or an integrated navigation menu.

Even though the survey in [ZZ16] is already a couple years old, the results are still very relevant today. When comparing visualizations mentioned in the study to those used in current media, we can see that the complexity has not increased significantly in the last ten years, even if the use of interactive infographics has expanded in online media. This is despite the development of more and more intricate visualizations by experts in this field. So it can be assumed that infographics in media are close to the limit of complexity for a regular reader to comprehend when reading a news article. To make sense of more uncommon types of visualizations some prior knowledge is needed in the field of visualization and often in the topic of the represented data. Again, this shows that not every visualization is equally useful to different target groups.

In addition to the various complexity of the visualization themselves there is also a difference in how much data is presented on a website. Since there is so much data available in the context of energy and there are so many factors to consider, some websites have a lot of detailed visualizations focusing on various topics, including production, consumption, renewables, prices, etc. Some examples of websites that are very information dense are [RS26], [Gan26], [RR25] or [Bun26]. While many of the visualizations are very informative and well made, it takes some time, effort and sometimes background knowledge to fully make sense of the visualizations and interpret them correctly. These kind of websites usually provide too much information for a casual reader that is only remotely interested in the topic. However, these sites are very important for experts and people who want to get a deeper understanding of the topic. Furthermore, all this data being available represented as visualizations is crucial for giving the public and journalists the possibility to fact check statements made in politics and media. They also provide reliable data from trusted sources in the respective areas, e.g. official german institutions like the *Statistisches Bundesamt*, *Umwelt Bundesamt*, the *Bundesnetzagentur* or institutes for national and international data collection, for example the *Frauenhofer Institut* or the *International Energy Agency*.

The article that will be used as basis for the tailored data stories is the one from [Zeiz6]. It is a good example of a data story that is targeted at the general public, which is the target group of the authors at *Die Zeit*, a big german newspaper. They use a lot of quite simple visualizations to cover different aspects of energy in Germany. Before every visualization they write some background information to put the data into context and explain the most important insight. Since the article is relatively long for a typical news story, it is divided into sections covering different topics. There is a small navigation menu at the top of the page to jump to the respective section. This allows the reader to quickly read the parts they are most interested in without having to search through the whole article. The section that will be used for the targeted data stories is the one about where the electricity in Germany comes from. The first visualization is a 100% stacked area chart to show the development of the different energy sources over the last month. This type of chart is not very common in general media, so it might be difficult for some people to understand the visualization and interpret the data correctly, especially when thinking about children and elderly. There is a hover interaction, letting the user highlight a specific energy source and see the exact values for each day. The same interaction can be done by hovering over the legend of the visualization. Another thing to note is that the colors used for the different energy sources resemble the colors a source of electricity is commonly associated with, e.g. yellow for solar, brown for coal or dark green for biomass. This makes the visualization more intuitive. The color coding is used consistently throughout the article, to avoid confusion when switching between the visualizations.

They use another stacked area chart showing the source of the electricity in 15 minute intervals over the last week, but in this graph the y-axis is not 100% like the previous chart. Instead it shows the values in MWh, highlighting that total electricity production is higher during the day because of solar power. It might be interesting to readers to put these numbers into perspective, for example by comparing them to electricity consumption. Often the consumption of a household or a city is used for this, another option would be the consumption in the same time span which would let users conclude the import/export of electricity. This could be done by extra text or by adding a line for consumption to the visualization, similar to the graph from [Gan26]¹. It is also interesting to note that the Zeit article uses MWh as a unit, while the production chart from [Gan26] uses MW. Since [Gan26], run by the *Fraunhofer Institut*, is the source of the data for the Zeit article it shows that they transferred the data from MW to MWh. The reason for this is probably that the general public is more familiar with the unit MWh, since they likely come in contact with it more often, e.g. in electricity bills or other media. MW is more often used by energy experts, who are part of the target audience for Energy-Charts. [Gan26] also simplified the categories of energy sources, for example by combining wind onshore and offshore or the different kind of coal into one category. This is a good example of simplifying a visualization for a wider audience.

¹<https://www.energy-charts.info/charts/power/chart.html?l=de&c=DE>

The third graph in this section is again a 100% stacked area chart, showing the development of the different energy sources over the last ten years. The data is available for every month. One of the most significant observations to note here, which is also included in the short text above the graph, is that nuclear energy is now included in the graph until 2023 when the last nuclear power plants shut down in Germany. Other things that can be seen here are the peaks of solar production in summer and a slight reduction of coal. The hover interaction is the same as in the previous two graphs, so the user should be familiar with it already. This chart is again not very easy to interpret, especially when there could be an interactive option to only show one or two energy sources at a time, which would make it easier to see the development of the selected energy source over time. Another option would be to let the user switch to a simpler visualization like a line chart showing the electricity production for every source over time, which would provide a better understanding of the development of the different energy sources in relation to each other.

The second to last chart shows the percentage of electricity production from renewable energy sources since 2015 per month as a bar chart. Again, the same hover interaction is used but since there is only one value per month the highlighting is not as useful. However, it still provides the exact value for the month, making it easier to read every data point. For the sake of compactness, this chart could be combined with the previous one by adding an option to only show renewable energy sources. It would add more complexity to the previous visualization but it would spare the reader the effort to think through another chart.

The last chart in the section shows the amount of battery capacity in Germany since 2015, again as a stacked area chart. The source of this chart is [Sau26]², a project by the University of Aachen. It includes values for large-scale batteries, industrial batteries and home batteries. The hover interaction is the same as in the previous charts. The original chart has a zoom function to enhance specific parts of the graph, which is not available in the Zeit article since this feature is not as useful on this rather simple graph and would likely be a confusing element. Another feature from the original graph not included in the article is the option to make the visualization fullscreen by clicking on a magnifying glass icon. This interaction however would be useful to include, especially for elderly people with visual impairments, to help them read the graph more easily. This could be added to every visualization on the site, since it is a simple interaction, that could also be explained in a short introduction in the beginning.

It has to be mentioned, that this analysis only includes interactive visualizations in online newspapers and websites providing data on the energy grid, even though the largest group of the population will probably be most familiar with static visualizations in other media. For example, infographics on social media could be relevant when considering children as a target group, while elderly people might be most familiar with graphics

²<https://battery-charts.de/de/home-de/>

shown in regular newspapers or on television. But since the focus of this thesis lies on interactive data stories these static visualizations are not included here. However, the use of data visualizations especially in social media is an interesting and developing field of research growing in relevance.

We can take away from this analysis which visualizations are commonly used in online media and what features regular news consumers might be familiar with. So using these as an orientation is a good starting point for creating a data story targeted at the general public. Also, some specific design features can be identified that might pose a challenge for PLA like missing tutorials for interactions (e.g. all visualizations that have a hover interaction), hardly readable text because of size or color (e.g. [RS26] has small, light gray text on white background) or difficult color schemes (e.g. the map of how fast different regions in Germany build renewables from [San26]) and the information density of some websites (e.g. [RS26], [Gan26], [RR25]). The information density is also a problem when considering children as a target group. Another topic that could be problematic for both children and older adults is the navigation of a website, as these target groups are usually less experienced with technology and the concept of using links to reach different chapters might not be familiar to them (e.g. [RS26], [Zeiz6]). There are some more conclusions that can be drawn from this analysis regarding children. The most prominent issue is a lack of engaging and colourful visuals in almost every visualization, which makes children lose interest quite quickly. Also, the content of the visualizations might be difficult to grasp as there is almost always some background knowledge required to understand it. As an example, children may not know how the different types of generation work and why it is important to focus on renewables. So more basic explanations are needed in this case.

4 | Implementation of prototypes

In this chapter the data stories for older adults and children will be implemented. The resulting visualizations and the choices made during the design process will be discussed.

The implemented stories will take inspiration from the article by *Die Zeit*, mentioned previously in [chapter 3](#). This is a good starting point, as it is published by a large german newspaper and targets the general public. It includes many visualizations that have potential to be tailored to older adults or children. Because the whole article is quite extensive, only the chapter about electricity generation called *Woher kommt der Strom?* (Where does the electricity come from?) will be looked at.

The first step of creating the data stories is to get the underlying data for the visualizations. In this case, the source of the data is the same as in the *Zeit* article, which is mainly the *Fraunhofer Institut*. It runs the website *Energy Charts* that was also examined in [chapter 3](#). This site offers a lot of data about different aspects of the German electricity market, like generation, price, load, energy trading, etc. This data is not only available for Germany but for all other European countries as well. All of it can be downloaded for free via an API in JSON format. For the data stories in this thesis, only data on German electricity generation is relevant, which is available in the categories *public_power* and *total_power*. The data for the graph on battery capacity comes from *Battery Charts*, a service provided by the University of Aachen, available at ¹. They provide a lot more detailed information regarding the use of batteries in the energy grid, most of it is too complex for the purpose of the data stories. The relevant data can be downloaded as a CSV file via a button next to the respective chart on their website.

Some data included in the download will not be necessary for the visualizations (e.g., electricity trading), so those data points will be filtered out. Also, some energy sources that are not relevant to the stories will be combined to simplify the visualizations. For example, Wind Onshore and Offshore will be combined into one category, or the different types of coal and gas will be merged into one category each. Some smaller energy sources, like waste, geothermal, and oil, will be added to the *other* category, since they are not relevant to the data stories and would add

¹<https://www.battery-charts.de>

unnecessary complexity to the visualizations. Since the provided data is already completely cleaned and well structured, no further initial data manipulation is required.

Like the *Zeit* article, the format of the stories is web-based, as it is the most common form to present such content. The most important tool for creating the data visualizations is D3.js, which is a powerful JavaScript library commonly used for creating interactive data visualizations. Parts of the code were generated with the help of AI tools, allowing for faster development and prototyping. The AI had no direct impact on the design decisions, the structure, the text or the overall content of the stories. Some data operations on the downloaded files were performed with the help of AI, mainly for aggregating the 15 minute data into daily or monthly summaries. Samples of the resulting numbers were manually checked to ensure that the intended calculations were correctly executed.

4.1 Older Adults

The data story for older adults will closely resemble the article from *Die Zeit*, since they are likely familiar with the newspaper's style and information presentation. Some adjustments will be made to improve readability and accessibility, such as using larger fonts, easily distinguishable colors, and adding explanations on how to interact with the visualizations. Also, some extra features are added to enhance the user experience for PLAs.

The tooltip that appears when hovering over the visualization will be slightly changed from the one used in the original article. Instead of only showing the value of the energy source that is currently being hovered over, it will display the values of all energy sources at that specific point in time. This avoids the required motor skills to move the mouse exactly over a specific layer, which can be quite challenging in the original version. This is especially true for energy sources that have smaller areas and are further up on the stacked charts, putting the bars for the different months at varying heights. This change also makes comparing the contributions of different energy sources much easier, since the exact number for every energy source is shown all at once and in order. The tooltip also has a larger font when compared to the original article, following the guiding principle of readability.

Staying on the topic of readability, the legend of the visualizations and the labels for the axis are also in a larger font. Since hovering over different energy sources in the legend highlights them in the visualization, this interaction will also be made easier when the legend is bigger. Also, by clicking on a category in the legend, one or more energy sources can be highlighted until the category is clicked again. This allows easier comparison between energy sources by hiding other categories. Because this feature might lead to some confusion for users, even with a short explanation, there is a button with the option to reset the highlighting and return to the original state of the chart. As a further accessibility feature, a button is added to each visualization making it appear in fullscreen. This

lets users view them in a larger format, which can be particularly helpful for people with visual impairments and makes it more comfortable to explore the visualization without distracting elements.

In the *Zeit* only area charts are used to visualize the data, which older adults might not be familiar with. In case they have trouble interpreting area charts, there is the added option to switch to a line chart instead. Apart from personal preference, the line chart also provides a better overview of individual trends for each category, making it easier to compare their variations over time. This is also an example of how the same data being visualized in different ways can lead to different insights, which might be interesting to see for people who are not experienced with visualizations. The option to switch between the two chart types is available for all visualizations to give users the most flexibility in how they want to view the data. The line chart offers the same interactive features as the area chart, so switching between them should feel intuitive.

The possible interactions for each visualization are explained when hovering over the i-button next to the headline of the chart. This gives people unfamiliar with interactive visualizations the chance to learn how to use them. Most interactions are used consistently throughout the story to keep the total number low and make each chart easier to navigate. Some charts include additional interactions, such as the option to highlight renewable energy sources or a time slider to adjust the time frame of the data shown in the chart. These interactions are mentioned in the text prior to the visualization, so users should be aware of them before interacting with the chart. They are also added to the i-button tooltip, so users can always check the available interactions for each chart.

To guide the user through the story, some text is added before each visualization to explain the data in the chart and to give some examples of insights that can be gained from it. The text is kept rather simple, avoiding too much technical detail. But the goal of the data story is to provide information that is hopefully interesting to PLAs and might give them a deeper understanding of German electricity generation beyond regular news articles. So there is a bit more explanatory text than the comparable section in the article from *Die Zeit*. However, since this story is only focused on one of the six original sections, the overall amount of information is still manageable. Giving examples of insights drawn from the data can encourage exploration by letting users try to find the corresponding pattern in the visualization themselves. This is especially important for older adults, who might not be used to analyzing data and gives them a starting point on how to reach their own conclusions.

In the first area chart, showing the daily electricity generation, the order of the layers is arranged in a way where energy sources with lower fluctuations are put at the bottom. This makes it easier to see the overall trends for the different energy sources, since the bars representing more stable sources will not be obscured by those with higher variability. Also, the colors for each energy source are those often associated with them, e.g., yellow for solar, green for biomass, and brown for coal. To keep the charts consistent, the same color scheme and layer order will be applied

across all visualizations. The first visualization is the simplest chart in the story that uses the data of the electricity generation over time, so it is a good starting point for users to get familiar with the general style and shared interactions of the visualizations. It is also a good way to introduce the line chart, since it provides a better overview of the individual percentages for each energy source, making them easier to compare.

The second visualization has a switch to put the focus on renewable energy sources. It reorders the layers so that renewable energy sources are placed at the bottom and adds a line for the total percentage of renewable energy in the overall energy mix. In the line chart, this automatically highlights all the renewable sources and adds the line for the total percentage of renewable energy. This allows users to easily see the contribution of renewables to the overall electricity generation and how it has changed over time. This visualization combines two of the graphs from the original article, where the total percentage of renewable sources is separate from the electricity mix over the years. It adds a bit of complexity to the chart, but it should still be understandable for older adults. And it removes the need to mentally combine the information from two separate charts, which can be quite difficult.

In the next graph, the total electricity generation is shown in a weekly overview. This chart has a time slider to adjust the time frame of the data shown in the chart. It also shows a line for the load, which is the total electricity consumption at that point in time. This lets users see how much electricity is being generated compared to how much is being consumed, and puts the absolute numbers in perspective. This is also not included in the *Zeit* article, but it is an interesting addition to the story, since absolute electricity generation numbers are not very meaningful on their own.

The last visualization shows how battery capacity in Germany has developed over the years. Compared to the other charts, this one is the simplest, since the data is not very complex. So it just has the standard interactions that are shared across all visualizations.

To add another customization option for the whole story, users can switch between light and dark mode based on their preference. The switch button between the modes is placed at the top of the page and remains accessible while scrolling through the data story so that it can be accessed at any time. For example, when users prefer reading in light mode but want to view the visualizations in dark mode, they can easily switch without losing their place in the story. The color scheme of the visualizations gets slightly changed as well, making colors easily distinguishable in both modes while staying close to the color associated with the energy source.

4.2 Children

The data story for children varies quite significantly from the basis of the *Zeit* article. Not only is the overall design more colorful and engaging, but the contents of the visualizations are simpler and more guided. It focuses

on only a few aspects of electricity generation, mainly the contribution of renewable energy sources over the years and the fluctuations of wind and solar energy throughout the seasons. To introduce children to the data story, some simple explanations are given about what renewable energy is in case they are not familiar with the concept. The text uses short sentences and simple language to make it easier to understand. Not much technical detail is included, to keep the focus on a few core concepts without overwhelming children with complex information.

To keep some structure in the story and make it somehow comparable to the *Zeit* article and the data story for older adults, each visualization is introduced with a short explanatory text that provides context and guides children on what to look for in the chart. The text also explains possible interactions with the chart and encourages to try them out. The overall length is kept relatively short to maintain the attention of the young audience. Therefore, the information about batteries is left out, since it takes a bit more understanding of the energy grid to fully grasp the role of electricity storage.

The first visualization compares the difference in production of solar and wind energy by using the least and most productive day of 2025 for each energy source. This is also a good way to make the data accessible to children, since they know from personal experience that some days are sunnier or windier than others. In order to avoid too much information input, the underlying data is very simplified by combining less relevant sources into broader categories, in this case fossil and other renewables. As children may have difficulties interpreting area charts, like the ones used in the *Zeit* article and the story for older adults, all the visualizations in this story are simple bar charts. With the abstract categorization, each bar only shows three categories: fossil, other renewables and the specific source being highlighted (solar or wind). This makes the chart easy to understand by focusing on one message. To make it more interactive, a slider is added so children can see changes in production between days. The slider allows them to move back and forth between the least and most productive days. An emoji is also added to each bar and adjusts in size based on the energy source's share. The color scheme is again based on colors mostly associated with the energy source.

The second chart shows the share of renewable sources, especially solar and wind energy, across the seasons. This expands the focus from individual days to a broader time frame by averaging the data for a whole month in each season. While this representation may not seem as engaging as the sliders and emojis in the first chart, it may still interest children to see the big differences between seasons. Furthermore, children can see the exact numbers by hovering over the bars, introducing them to a widely used interaction concept. This shifts the focus to comparing the numbers and shows how the natural assumption of more sunshine in summer is reflected in the data.

The third and final chart builds on the previous visualization by expanding the time frame once again to show changes in the share of renewable energy sources over multiple years. This is meant to show Germany's

transition from fossil fuels to renewable energy in its overall electricity generation mix. It keeps the theme of seasons by showing differences in renewable energy contributions within each season over the years. Besides the hovering interaction like in the last chart, there are buttons to switch between different seasons, allowing children to focus on one season at a time and observe how the share of renewable energy changes over the years. It can be seen that the share of fossil fuels decreases regardless of the season, while the share of renewable energy sources correspondingly increases over multiple years. The severity of this effect varies by season, with summer and spring showing the largest changes due to solar energy.

5 Conclusion and Future Work

There are definitely some logical design decisions that can be made to improve data visualizations for different target audiences, like children and older adults. One thing that is a very important consideration for both children and older adults is the amount and the complexity of information presented. Understanding the cognitive limitations of these target groups is essential to not overwhelm them while still presenting insights they might not have known otherwise. However, this line is difficult to navigate because of the differences within the groups, which also makes researching the effectiveness of different design choices challenging.

Most current visualizations targeted at the general population lack some features that would make them more accessible for older adults, such as larger fonts, higher contrast, simplified layouts, and clear instructions for possible interactions. Some of these features could be incorporated into existing visualizations, especially accessibility options like adjustable font sizes or easily distinguishable color schemes. Since accessibility is not only a matter for older adults, this should be a widespread standard in more visualizations aimed at the general population.

Since children do not often come in contact with news articles like the ones from *Die Zeit*, it is understandable that the visualizations are not designed with children as a target audience. An important consideration for children is how to get and keep their attention throughout the data story and ensure that they can learn from it. The most crucial aspect when it comes to data visualizations for children is to make sure they understand what is shown and explain how the data is represented. Because at this age improving the visualization literacy is often more important than the content itself, which helps them correctly interpret visualizations in the future.

The biggest limitation of this thesis is that the implemented data stories are only a result of theories and concept ideas that seem promising to reach the targeted audiences. Empirically evaluating the effectiveness of the proposed data stories would require a comprehensive survey among children and older adults. It would be interesting to present the data stories to members of the respective target groups and get their feedback on usability, understanding and overall experience. Especially a comparison between the article from *Die Zeit* that the stories are based on and the resulting data stories would provide valuable insights on whether the

proposed stories actually improve comprehension and engagement. To expand this idea further, multiple different versions of the data stories could be created with different designs and interaction possibilities to test the effectiveness of different design principles.

To give an example, it would be interesting to know whether older adults prefer the combination of the chart that shows the percentage of renewable energy and the chart that shows the percentage of each energy source individually or if they would prefer to see them separately. Combining these charts makes the single chart more complicated and potentially harder to understand. But when only looking at the individual sources of energy, an intuitive question might be how high the percentage of renewables is. With the combined chart the audience can get an instantaneous answer to this, whereas using separate charts require them to read further and mentally combine the information to get the same insight. Another question could be whether adding the timeslider and a line for the load in the weekly graph are perceived as interesting elements or if they are more confusing than helpful. This is something that is difficult to tell based on theoretical findings and could best be determined through user testing and feedback.

The same applies to the data story for children, where it would be intriguing to see to what extent the inclusion of interactive elements like sliders and emojis affects engagement and understanding. Whether there should be more interactive elements and at what point they might become distracting is another aspect worth exploring. Other factors to investigate would be how much text children are willing to read before they lose interest or if children are more likely to learn from the data story when they are guided through it by an adult or teacher, compared to exploring it independently.

However, developing different versions of the data stories and comparing them in a controlled study would require a significant amount of time and effort and depends on participation from members of the respective target groups, which can be difficult to achieve. Therefore, the effectiveness of the implemented data stories in this thesis remains largely theoretical.

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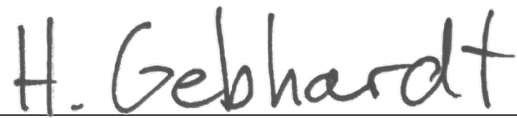
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Declaration of Authorship

Ich erkläre hiermit gemäß § 17 Abs. 2 APO, dass ich die vorstehende Bachelorsarbeit selbständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Bamberg, den 29.09.2026



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